Multiscale Spiking Graph Convolution Aggregation Network for Hyperspectral Image Classification

Shufang Xu, *Member, IEEE*, Wenxuan Fei, Gaoxu Wang, Runhua Sheng, *Graduate Student Member, IEEE*, Zhonghao Chen, *Graduate Student Member, IEEE*, and Hongmin Gao, *Senior Member, IEEE*

***Abstract*—Hyperspectral image (HSI) classification remains challenging due to complex spectral–spatial patterns, the growing demand for energy-efficient modeling in resource-constrained remote sensing applications, and the structural limitations of existing graph convolutional network (GCN)- and spiking neural network (SNN)-based methods. Conventional GCNs rely on static graphs and coarse neighborhood aggregation, limiting their ability to model non-Euclidean spectral–spatial relations, while most SNN-based approaches exploit event-driven sparsity for low-energy computation but lack effective adaptive multiscale and structured relational modeling. To address these issues, we propose a multiscale spiking graph convolution aggregation network (MS2GCAN) that integrates spike-driven temporal dynamics with adaptive graph learning. MS2GCAN comprises a multiscale temporal enhancement module for fine-to-coarse spatial encoding, an adaptive spiking graph convolution (ASGC) module with feature-dependent dynamic graphs, and a center spatiotemporal aggregation (CSTA) module for multihop spatial fusion with learnable temporal weighting. Experiments on three public hyperspectral datasets demonstrate that MS2GCAN achieves state-of-the-art (SOTA) performance with only 0.06–0.18 M parameters. In addition, the proposed model exhibits sparse spiking activity across layers, indicating strong potential for energy-efficient computation. These results validate the effectiveness of MS2GCAN for accurate and resource-efficient HSI classification (HSIC). The source code will be released at https://github.com/fwx0314/MS2GCAN**

***Index Terms*—Dynamic graphs, graph convolutional network (GCN), hyperspectral image (HSI) classification, multiscale, spiking neural network (SNN).**

I. Introduction

HYPERSPECTRAL images (HSIs) [1] offer fine-grained spectral measurements across hundreds of contiguous bands, enabling precise material identification for various remote sensing tasks [2], [3], including object detection [4], [5], classification [6], [7], and change detection [8], [9]. Recently, the emergence of large-scale geospatial datasets, such as MEET [10], has further highlighted the demand for fine-grained scene classification across diverse scales, underscoring the need for models to handle complex spatial heterogeneities and subtle class distinctions effectively. Yet, their high dimensionality, strong spectral redundancy, and spatial heterogeneity continue to pose key challenges, such as spectral confusion [11], [12], spectral variability [13], [14], and the Hughes phenomenon [15], [16].

HSI classification (HSIC) aims to categorize each pixel by jointly interpreting spectral and spatial information [17]. Traditional classification methods, such as support vector machines (SVMs) [18], random forests (RFs) [19], k-nearest neighbors (KNNs) [20], and sparse representation methods [21], primarily rely on spectral signatures and thus overlook crucial spatial context. While principal component analysis (PCA) [22] and linear discriminant analysis (LDA) [23] reduce redundancy, their handcrafted design restricts adaptability to complex scenes.

Deep learning has significantly advanced HSIC [24]. Convolutional neural network (CNN)-based models [25], [26], [27], [28], [29] capture spectral–spatial patterns but remain inherently limited by local receptive fields, making long-range context modeling difficult. Transformer-based methods [30], [31], [32] introduce global attention, yet they often sacrifice local precision, rely on large attention blocks, and require substantial computation.

Graph convolutional networks (GCNs) offer a complementary modeling strategy by representing pixels or superpixels as nodes and explicitly capturing non-Euclidean spectral–spatial relations. Existing approaches [33], [34], [35] enhance multiscale aggregation and structural encoding, but two fundamental limitations persist: 1) static or manually designed graphs that cannot adapt to evolving feature representations and 2) coarse multihop aggregation, where information from different hops is often fused with fixed or uniform weighting, limiting fine-grained spatial reasoning.

Despite these advances, most deep HSIC models are implemented as artificial neural networks (ANNs), whose dense computations and continuous-valued activations often lead to considerable computational and energy overhead [36]. This significantly restricts their deployment on resource-constrained edge platforms, such as uncrewed aerial vehicles (UAVs) and satellites [37]. Balancing classification accuracy and deployment efficiency therefore remains a fundamental challenge.

Received 26 January 2026; revised 8 March 2026; accepted 20 March 2026. Date of publication 27 March 2026; date of current version 16 April 2026. This work was supported by the Basic Research Program of Jiangsu under Grant BK20252043. *(Corresponding author: Hongmin Gao.)*

Shufang Xu, Wenxuan Fei, and Hongmin Gao are with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: xushufang@hhu.edu.cn; 241607010003@hhu.edu.cn; gaohongmin@hhu.edu.cn).

Gaoxu Wang is with the National Key Laboratory of Water Disaster Prevention, Nanjing Hydraulic Research Institute, Nanjing 210029, China, and also with the Yangtze River Conservation and Green Development Institute, Nanjing 210098, China (e-mail: gxwang@nhri.cn).

Runhua Sheng is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: shengrunhua@whu.edu.cn).

Zhonghao Chen is with the College of Information Science and Engineering, Hohai University, Nanjing 211100, China (e-mail: chenzhonghao@hhu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3678343

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Spiking neural networks (SNNs) provide a promising direction for high-performance and energy-efficient HSIC modeling [38]. On one hand, SNNs offer significant energy efficiency by replacing dense multiply–accumulate (MAC) operations with sparse, event-driven addition-accumulation (AC) operations, facilitating deployment on resource-constrained UAV or satellite platforms [39]. On the other hand, the spiking threshold mechanism acts as an inherent nonlinear denoiser to suppress sensor noise, while its temporal integration functions as a dynamic filter that accumulates salient features over time to mitigate spectral variability. Thus, SNNs provide a robust and efficient framework for modeling high-dimensional spectral–spatial data in complex remote sensing scenes [40].

Recent SNN-based HSIC methods [38], [41], [42], [43], [44], [45] demonstrate the potential of spiking models, yet most architectures remain heavily convolution-driven and inherit CNNs' locality bias. Moreover, achieving competitive accuracy often requires wide multibranch designs or long simulation time steps, leading to increased architectural and temporal complexity. Critically, existing SNN-based approaches generally lack mechanisms for adaptive multiscale spatial encoding and non-Euclidean graph-based relational modeling, limiting their representational capacity in complex and heterogeneous scenes.

However, when viewed holistically, existing methods still exhibit three critical and interconnected gaps in high-performance and energy-efficient HSIC modeling.

1) Existing GCN-based models lack the intrinsic temporal dynamics of spiking networks and do not support feature-dependent dynamic graph construction, leading to static or weakly adaptive spatiotemporal representations.
2) Current GCN frameworks rely on coarse neighborhood aggregation and cannot achieve learnable, fine-grained multihop fusion on the target pixel.
3) Existing SNNs for HSIC lack lightweight modules for explicit, adaptive multiscale spatial encoding and structured non-Euclidean graph modeling, limiting their ability to capture comprehensive fine-to-coarse patterns.

To bridge these gaps, we propose an MS2GCAN. Unlike existing lightweight CNNs restricted to static Euclidean grids or convolution-driven SNNs that overlook structured dependencies, MS2GCAN uniquely integrates spiking temporal dynamics with adaptive graph learning. The main contributions are summarized as follows.

1) We propose MS2GCAN, the first unified framework bridging SNNs and GCNs for HSIC. It synergizes SNNs' energy efficiency and denoising with GCNs' structural modeling by embedding non-Euclidean reasoning into the spiking temporal domain.
2) We propose a multiscale temporal enhancement (MSTE) module for adaptive multiscale spatial feature encoding in the spiking domain, based on parallel depthwise separable convolutions and a gated fusion mechanism.
3) We introduce an adaptive spiking graph convolution (ASGC) module with an adaptive Pearson adjacency (APA) mechanism, enabling feature-dependent dynamic graph construction and non-Euclidean spectral–spatial relational modeling.
4) We design a center spatiotemporal aggregation (CSTA) module that facilitates fine-grained spatiotemporal fusion through learnable hopwise spatial weighting and spike-driven temporal weighting.

The remainder of this article is organized as follows. Section II reviews related work on SNNs and GCNs for HSIC. Section III details the proposed MS2GCAN architecture. Section IV presents experimental results and analysis. Finally, Section V concludes the article and discusses future work.

II. Related Work

This section reviews recent advancements in SNNs and GCNs for HSIC.

A. SNNs for HSIC

SNNs have recently attracted growing attention for HSIC due to their event-driven computation and inherent energy efficiency [38], [41]. Existing SNN-based HSIC methods focus on improving spatiotemporal representation and classification accuracy under resource constraints through architectural and training innovations.

Liu et al. [38] introduced SNN-SSEM, which integrates shuffle squeeze-and-excitation modules and channel-shuffle attention into spiking networks for HSIC. Building on efficient spiking representations, Liu et al. [41] proposed a spiking architecture based on LIF neurons and depthwise separable convolutions, namely SNN-DP, employing direct spike encoding to facilitate efficient spatiotemporal feature learning. To enhance boundary discrimination and alleviate spectral confusion, Wang et al. [42] introduced BDSNN, incorporating deformable convolutions and boundary-aware mechanisms into spiking architectures. Focusing on deployment efficiency, Liu et al. [43] developed a knowledge distillation-based lightweight spiking framework that significantly reduces inference latency and model complexity. More recently, Li et al. [44] explored adaptive self-attention and refocusing convolution to capture longer-range spatial dependencies within limited simulation timesteps, while Li et al. [45] proposed FRM-SNN, designing a multibranch residual spiking architecture for richer spatial context aggregation.

Despite these advances, most existing SNN-based methods remain predominantly convolution-driven and inherit the locality bias of CNNs. Their spatial modeling is largely restricted to fixed receptive fields, making it difficult to capture adaptive multiscale structures and structured non-Euclidean spectral–spatial relations. Moreover, achieving competitive accuracy often relies on widened multibranch designs or increased temporal depth, which partially offsets the efficiency advantages of spiking computation. These limitations indicate that effectively integrating spike-driven temporal dynamics with adaptive non-Euclidean spatial modeling remains an open challenge for efficient and accurate HSIC.

B. GCNs for HSIC

GCNs have been widely adopted for HSIC by representing pixels or superpixels as graph nodes and modeling

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spectral–spatial relations in a non-Euclidean manner [46]. Existing GCN-based HSIC methods mainly differ in graph construction strategies and neighborhood aggregation schemes.

Early methods typically construct superpixel- or region-level graphs to reduce computational complexity and incorporate long-range contextual information. Wan et al. [47] proposed CAD-GCN, projecting pixels into regional nodes and employing dynamic edge filtering to enhance nonlocal interactions. Liu et al. [48] proposed CEGCN, introducing a graph encoder–decoder framework to jointly model pixel-level and superpixel-level representations. Ding et al. [49] proposed MFGCN, integrating multiscale CNN features with superpixel-based graph reasoning for improved spectral–spatial fusion. Wang et al. [50] proposed LG-GCN, adopting a lightweight dual-branch architecture to capture both local and global dependencies with reduced computational overhead. Zhou et al. [35] proposed AMGCFN, incorporating parallel multiscale convolutions and multihop graph propagation to enhance hierarchical representation. Liu et al. [51] proposed HDECGCN, refining graph representations via adaptive adjacency weighting. Despite their effectiveness, these superpixel- or region-level approaches rely on global graph modeling and offline training, which limits flexibility when handling newly arriving samples.

To address this limitation, pixel-level graph construction methods based on patch subgraphs were introduced. Hong et al. [33] proposed miniGCN, performing training on sampled subgraphs to enable efficient optimization without retraining the entire network. Zhang et al. [52] proposed NAGIN, introducing a neighborhood-adaptive mechanism to dynamically refine graph connections based on local spectral characteristics. Yang et al. [53] proposed GTFN, integrating Transformer-based attention into graph learning to jointly capture spectral sequences and spatial dependencies. In our previous works, we proposed MRCAGCFN [54] and MS2FN [55], which further explored multiscale convolutional and graph-based fusion strategies for adaptive spectral–spatial representation.

However, most existing GCN-based HSIC models still rely on static or quasi-dynamic graphs constructed from ANN-based features, fundamentally limiting their ability to adapt to rapidly evolving representations during training. Moreover, node aggregation is often coarse and single-scale, with multihop information fused using fixed or uniform weighting schemes, making it difficult to prioritize the most discriminative cues for the central pixel. Most importantly, existing GCN-based HSIC models lack intrinsic spike-driven temporal dynamics, restricting their capacity to efficiently capture fine-grained spatiotemporal correlations. These observations suggest that jointly enabling dynamic, feature-dependent graph construction and fine-grained spatiotemporal aggregation remains insufficiently explored in existing GCN-based HSIC methods.

III. Proposed Method

A. Preliminaries

1) Spiking Neuron Model: We adopt a standard discrete-time leaky integrate-and-fire (LIF) neuron implemented using an existing SNN framework [56]. At each timestep t, the membrane potential of neuron i is updated as follows:

$$v_i^{(t)} = \beta v_i^{(t-1)} + \sum_j W_{ij} s_j^{(t-1)} \tag{1}$$

where β denotes the membrane decay factor, W_{ij} is the synaptic weight from neuron j to neuron i, and $s_j^{(t-1)} \in \{0, 1\}$ is the spike output of neuron j at the previous timestep. A hard reset is applied after firing. Standard parameter settings from the framework are used unless otherwise specified. The neuron dynamics are not the focus of this work; LIF neurons serve as computational primitives for spatiotemporal feature modeling.

2) Surrogate Gradient Method: The nondifferentiable spike generation prevents direct gradient backpropagation. We adopt the surrogate gradient method, replacing the derivative of the spike function with a smooth approximation during backpropagation while keeping the forward spiking unchanged. Specifically, the gradient is approximated using an arctangent surrogate

$$\frac{\partial s_i^{(t)}}{\partial v_i^{(t)}} \approx \frac{1}{1 + \left[\alpha \left(v_i^{(t)} - \theta\right)\right]^2} \tag{2}$$

where $\alpha > 0$ controls the smoothness of the approximation.

B. Overview of Proposed MS2GCAN

The overall MS2GCAN architecture for HSIC is illustrated in Fig. 1(a). An input patch $\mathbf{X} \in \mathbb{R}^{B\times L\times L}$ is first repeated T times to form a spatiotemporal sequence $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{T\times B\times L\times L}$. Theoretically, this "pseudotemporal" repetition serves as a mechanism for stochastic feature refinement via temporal evidence accumulation. Unlike deterministic single-pass networks, SNNs treat the temporal dimension as a domain for statistical integration. By repeating the static input, the LIF neurons progressively accumulate membrane potentials, effectively acting as a low-pass filter that suppresses transient spatial noise inherent in HSI sensors. This iterative process allows the network to reinforce salient spectral–spatial features through successive firing thresholds, transforming static intensities into robust, spike-based representations that are more resilient to intraclass variability.

The sequence $\mathbf{X}_{\text{spa}}$ is first processed by the spectral transformation module (STM) to compress spectral redundancy and convert features into spike representations. The resulting spike sequence is then fed into three synergistic modules for spatiotemporal feature learning. The MSTE module captures fine-to-coarse spatial patterns via parallel depthwise separable spiking convolutions. A lightweight gating network adaptively fuses multiscale features at each timestep, producing spatially enriched spike representations. ASGC constructs timestepwise dynamic non-Euclidean graphs using pairwise Pearson correlations and performs spike-driven graph convolution with LIF neurons, enabling iterative, pseudotemporal neighborhood aggregation. Finally, CSTA integrates information from the center node and its multihop neighbors. A gating network generates K-hops' weights for spatial fusion, while learnable temporal weights aggregate membrane potentials across timesteps, producing the center node's integrated spatiotemporal representation.

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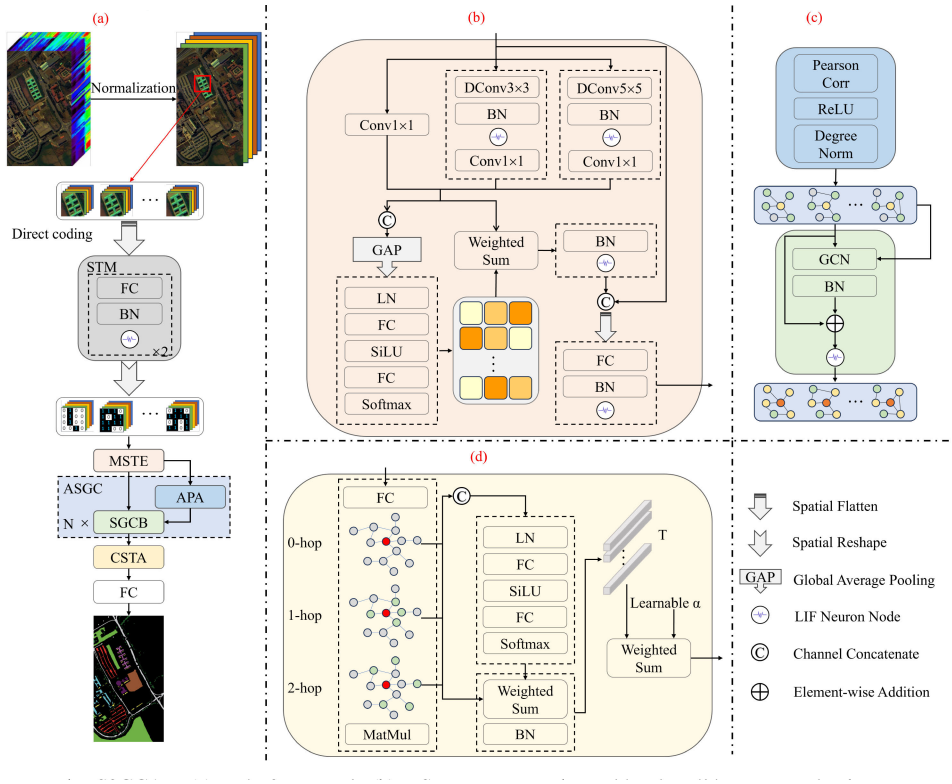


Fig. 1. Schematic of our proposed MS2GCAN. (a) Main framework. (b) MSTE represents the multiscale spiking temporal enhancement. (c) ASGC, representing the adaptive spiking graph convolution, consisted of APA and SGCB. APA represents the adaptive Pearson adjacency and SGCB represents the spiking graph convolution block. (d) CSTA denotes the center spatiotemporal aggregation module.

The aggregated features are fed into a fully connected (FC) layer for class prediction, optimized with cross-entropy loss

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log\left(\hat{y}_{i,c}\right) \tag{3}$$

where N and C denote the number of samples and classes, respectively, yi, c is the ground-truth label, and $\hat{y}_{i,c}$ is the predicted probability. Furthermore, while the repetition increases the number of timesteps, the event-driven nature and binary sparsity of SNNs ensure that the overall computational energy remains significantly lower than that of dense floating-point matrix multiplications in conventional GCNs. This architecture effectively combines the iterative feature propagation capability of SNNs with the structural reasoning of GCNs, yielding enhanced representation power for static hyperspectral patches.

C. Proposed STM

STM is designed to compress spectral redundancy and extract discriminative spectral features from HSIs, and its structure is illustrated in Fig. 1(a). The module takes $\mathbf{X}_{\text{spa}}$ as input, which is first spatially flattened into $\mathbf{X}_{\text{seq}} \in \mathbb{R}^{T\times L^2\times B}$. The sequence is processed by a two-layer structure, each consisting of an FC transformation, batch normalization (BN), and an LIF spiking neuron

$$\mathbf{X}_{\text{seq}}^{(1)} = \text{LIF}\left(\text{BN}\left(\text{FC}_{B\to 2-D}\left(\mathbf{X}_{\text{seq}}\right)\right)\right) \tag{4}$$

$$\mathbf{X}_{\text{seq}}^{(2)} = \text{LIF}\left(\text{BN}\left(\text{FC}_{2-D\to D}\left(\mathbf{X}_{\text{seq}}^{(1)}\right)\right)\right) \tag{5}$$

where $\mathbf{X}_{\text{seq}}^{(1)} \in \mathbb{R}^{T\times L^2\times(2-D)}$ denotes the intermediate spike-encoded spectral features, $\mathbf{X}_{\text{seq}}^{(2)} \in \mathbb{R}^{T\times L^2\times D}$ represents the final output with D being the target spectral dimension, FC(·) denotes the FC layer, BN(·) represents the BN layer, and LIF(·) denotes the LIF spiking activation layer. The first FC layer [see (4)] also serves as the spike encoding layer.

D. Proposed MSTE Module

The MSTE module addresses the limitation of fixed receptive fields in capturing the diverse spatial scales of heterogeneous land covers. It adaptively integrates multi-scale spatial features via parallel depthwise separable spiking convolutions with a learnable gating mechanism. Given the spectral-transformed sequence $\mathbf{X}_{\text{seq}}^{(2)} \in \mathbb{R}^{T\times L^2\times D}$ from STM, it is reshaped into a 2-D spatial layout $\mathbf{X}'_{\text{spa}} \in \mathbb{R}^{T\times D\times L\times L}$ for spatial feature extraction [see Fig. 1(b)].

To capture spatial dependencies across multiple scales, ranging from local pixel-level (1×1) to regional-level (5×5) granularity, three parallel convolutional branches are employed

$$\mathbf{X}_{\text{spa}}^{(1)} = \text{Conv2d}_{1\times 1}\left(\mathbf{X}'_{\text{spa}}\right) \tag{6}$$

$$\mathbf{X}_{\text{spa}}^{(2)} = \text{Conv2d}_{1\times 1}\left(\text{LIF}\left(\text{BN}\left(\text{DConv2d}_{3\times 3}\left(\mathbf{X}'_{\text{spa}}\right)\right)\right)\right) \tag{7}$$

$$\mathbf{X}_{\text{spa}}^{(3)} = \text{Conv2d}_{1\times 1}\left(\text{LIF}\left(\text{BN}\left(\text{DConv2d}_{5\times 5}\left(\mathbf{X}'_{\text{spa}}\right)\right)\right)\right) \tag{8}$$

where $\mathbf{X}_{\text{spa}}^{(i)} \in \mathbb{R}^{T\times D\times L\times L}$ is the ith scale feature map, and Conv2d(·) and DConv2d(·) denote pointwise and depthwise convolutions, respectively.

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The generated multiscale features are adaptively fused via a lightweight gating network. The branch outputs are concatenated and spatially pooled to yield a scale representation

$$\mathbf{M}_{\text{cat}} = \text{Concat}\left[\mathbf{X}_{\text{spa}}^{(1)}, \mathbf{X}_{\text{spa}}^{(2)}, \mathbf{X}_{\text{spa}}^{(3)}\right] \in \mathbb{R}^{T\times 3-D\times L\times L} \tag{9}$$

$$\mathbf{M}_{\text{mean}} = \text{Mean}_{(L,L)}\left(\mathbf{M}_{\text{cat}}\right) \in \mathbb{R}^{T\times 3-D} \tag{10}$$

where $\text{Mean}_{(L,L)}(\cdot)$ denotes global pooling in spatial dimension, and $\mathbf{M}_{\text{mean}}$ is the scale representation. The gating network computes the scalewise fusion weights

$$\mathbf{W}_{\text{ms}} = \text{Softmax}\left(\text{FC}_{D\to 3}\left(\text{SiLU}\left(\text{FC}_{3-D\to D}\left(\text{LN}\left(\mathbf{M}_{\text{mean}}\right)\right)\right)\right)\right) \tag{11}$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\text{SiLU}(\cdot)$ is the intermediate activation function, and Softmax normalizes the weights across the three scales. The resulting $\mathbf{W}_{\text{ms}} \in \mathbb{R}^{T\times 3}$ provides timestepwise, scale-aware fusion weights, allowing the model to adaptively focus on different spatial scales depending on the object's size and scene complexity.

The fused multiscale representation is then computed as a weighted sum

$$\mathbf{M} = \sum_{i=1}^{3} \mathbf{w}_i \odot \mathbf{X}_{\text{spa}}^{(i)} \tag{12}$$

where $\mathbf{w}_i$ is the ith weight vector extracted from $\mathbf{W}_{\text{ms}}$, $\odot$ denotes elementwise multiplication, and $\mathbf{M}$ represents the fused multiscale feature. The fused output is normalized and passed through LIF

$$\mathbf{M}_{\text{out}} = \text{LIF}\left(\text{BN}\left(\mathbf{M}\right)\right). \tag{13}$$

Finally, $\mathbf{M}_{\text{out}}$ is flattened to $\mathbf{M}_{\text{seq}} \in \mathbb{R}^{T\times L^2\times D}$ and concatenated with the original spectral features $\mathbf{X}_{\text{seq}}^{(2)}$

$$\mathbf{X}_{\text{cat}} = \text{Concat}\left[\mathbf{X}_{\text{seq}}^{(2)}, \mathbf{M}_{\text{seq}}\right] \in \mathbb{R}^{T\times L^2\times 2-D}. \tag{14}$$

The final output is obtained through an FC–BN–LIF projection

$$\mathbf{X}_{\text{ms}} = \text{LIF}\left(\text{BN}\left(\text{FC}_{2-D\to D}\left(\mathbf{X}'_{\text{seq}}\right)\right)\right). \tag{15}$$

The MSTE module thus produces a spatially enriched representation $\mathbf{X}_{\text{ms}}$, integrating spectral information with adaptively fused multiscale spatial features across simulated timesteps.

E. Proposed ASGC Module

To address the limitations of static graphs in capturing dynamic, non-Euclidean spectral–spatial correlations, we propose the ASGC module. It constructs timestepwise dynamic graphs via APA, enabling modeling of evolving non-Euclidean relationships in the spiking domain [see Fig. 1(c)]. The input $\mathbf{X}_{\text{ms}}$ from MSTE provides spatially enriched spike features for graph construction. ASGC captures non-Euclidean dependencies through two key components: the APA matrix and the spiking graph convolution block (SGCB).

1) Adaptive Pearson Adjacency: To capture feature-dependent relationships in spiking representations, ASGC module constructs a dynamic adjacency matrix based on Pearson correlation coefficients. Unlike Euclidean or Cosine metrics that are sensitive to absolute spectral intensity or baseline shifts, the Pearson-based approach emphasizes relative structural correlations, which is particularly suited for spiking dynamics.

The adjacency is computed from the multiscale spike features $\mathbf{X}_{\text{ms}}$. We first perform feature centering to mitigate the risk of membrane potential saturation, where absolute high-intensity spectral bands might otherwise dominate the spiking activity regardless of their discriminative value

$$\mu = \text{Mean}_{(D)}\left(\mathbf{X}_{\text{ms}}\right) \tag{16}$$

$$\mathbf{X}_{\text{center}} = \mathbf{X}_{\text{ms}} - \mu \tag{17}$$

where $\mathbf{X}_{\text{center}}$ represents the zero-centered feature. By subtracting the mean, APA filters out common-mode noise and background offsets, ensuring that the potential accumulation in subsequent spiking neurons is driven by intrinsic spectral variations rather than absolute energy levels. Subsequently, L2-normalization is applied to obtain standardized relational vectors

$$\mathbf{X}_{\text{norm}} = \frac{\mathbf{X}_{\text{center}}}{\|\mathbf{X}_{\text{center}}\|_2}. \tag{18}$$

From the perspective of spiking neuron dynamics, the centering operation stabilizes membrane potential accumulation in LIF neurons. The membrane potential evolves as follows:

$$V_i\left(t+1\right) = \lambda V_i\left(t\right) + I_i\left(t\right), \quad I_i\left(t\right) = \sum_j w_{ij} x_j\left(t\right). \tag{19}$$

If the features contain a global bias b, i.e., $x_j(t) = \tilde{x}_j(t) + b$, the input current becomes $I_i(t) = \sum_j w_{ij}\tilde{x}_j(t) + b\sum_j w_{ij}$, introducing a constant drift in the membrane potential. By centering the features ($x'_j = x_j - \mu$), the expected input current satisfies $\mathbb{E}[I_i(t)] \approx 0$, preventing bias-driven accumulation and ensuring that spike firing is primarily driven by relative feature variations. This property aligns naturally with the Pearson correlation formulation used in APA.

The pairwise similarity and the normalized dynamic adjacency matrix are then derived as follows:

$$\mathbf{S} = \text{ReLU}\left(\mathbf{X}_{\text{norm}}\mathbf{X}_{\text{norm}}^{\top}\right) \in \mathbb{R}^{T\times L^2\times L^2} \tag{20}$$

$$\mathbf{D} = \text{diag}\left(\sum_j \mathbf{S}_{ij} + \epsilon\right), \quad \mathbf{A}_{\text{dynamic}} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} \tag{21}$$

where $\text{ReLU}(\cdot)$ ensures nonnegative adjacency values. This centering-and-normalization pipeline aligns with the event-driven nature of SNNs, as it encourages stable and sparse spiking activity by focusing on relative feature-driven dependencies. This allows the spiking graph convolution to model flexible, non-Euclidean relationships while preserving the representational precision of the spiking domain.

2) Spiking Graph Convolution Block: Given the dynamic adjacency $\mathbf{A}_{\text{dynamic}}$ and initial node features $\mathbf{H}^{(0)} = \mathbf{X}_{\text{ms}}$, the lth spiking graph convolution layer is

$$\mathbf{H}^{(l+1)} = \text{LIF}\left(\text{BN}\left(\mathbf{A}_{\text{dynamic}}\left(\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right)\right) + \mathbf{H}^{(l)}\right) \tag{22}$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{D\times D}$ is learnable. Residual connections facilitate feature reuse and stable training, while BN and LIF neurons normalize and nonlinearly transform features, enabling spike-driven graph convolution across feature-dependent neighborhoods.

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By jointly learning the graph topology and performing multistep propagation, ASGC captures long-range, non-Euclidean relationships among nodes, producing integrated spatiotemporal representations for HSI feature learning.

F. Proposed CSTA Module

The CSTA module enhances pixelwise classification by adaptively integrating multihop neighborhood context and learnable temporal weighting, balancing the center pixel's spectral fidelity with spatial information [see Fig. 1(d)]. Given spiking features $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{T\times L^2\times D}$ produced by the ASGC module and the adaptive adjacency $\mathbf{A}_{\text{dynamic}}$, CSTA generates discriminative center-node representations by integrating multihop spatial aggregation with learnable temporal fusion.

Features from the center pixel up to K-hop neighbors are aggregated iteratively. Let the k-hop adjacency matrix be

$$\mathbf{A}^{(k)} = \begin{cases} \mathbf{I}, & k = 0 \\ \mathbf{A}_{\text{dynamic}}\mathbf{A}^{(k-1)}, & 1 \le k \le K \end{cases} \tag{23}$$

where $\mathbf{I}$ is the identity matrix. Let $\mathbf{a}^{(k)}$ denote the row of $\mathbf{A}^{(k)}$ corresponding to the center node. The normalized hopwise connectivity $\mathbf{m}^{(k)}$ is computed as follows:

$$\mathbf{m}^{(k)} = \frac{\mathbf{a}^{(k)}}{\sum_j \mathbf{a}_j^{(k)} + \epsilon}. \tag{24}$$

The aggregated feature of the kth hop, $\mathbf{c}^{(k)}$, is a weighted combination of the FC layer enhanced node features

$$\mathbf{c}^{(k)} = \mathbf{m}^{(k)}\text{FC}_{D\to D}\left(\mathbf{X}_{\text{gcn}}\right). \tag{25}$$

The hopwise features $\mathbf{c}^{(k)} \in \mathbb{R}^{T\times D}$ are concatenated

$$\mathbf{C}_{\text{cat}} = \text{Concat}\left[\mathbf{c}^{(0)}, \ldots, \mathbf{c}^{(K)}\right] \in \mathbb{R}^{T\times(K+1)D}.$$

A lightweight gating network generates hopwise attention weights $\mathbf{W}_{\text{hops}} \in \mathbb{R}^{T\times(K+1)}$. To maintain architectural simplicity, the gating module is implemented as a nonspiking lightweight ANN head (FC layers with SiLU activation). It estimates the relative importance of different hop features but does not involve membrane potential dynamics. The resulting weights are applied to the aggregated spiking features $\mathbf{c}^{(k)}$, thereby preserving the spike-driven nature of the core feature representation in MS2GCAN. The formal computation of $\mathbf{W}_{\text{hops}}$ is defined as follows:

$$\mathbf{U} = \text{SiLU}\left(\text{FC}_{(K+1)D\to D}\left(\text{LN}\left(\mathbf{C}_{\text{cat}}\right)\right)\right) \tag{26}$$

$$\mathbf{W}_{\text{hops}} = \text{Softmax}\left(\text{FC}_{D\to K+1}\left(\mathbf{U}\right)\right). \tag{27}$$

The final fused center feature $\mathbf{C}_{\text{fused}} \in \mathbb{R}^{T\times D}$ is obtained via the weighted sum and BN

$$\mathbf{C}_{\text{fused}} = \text{BN}\left(\sum_{k=0}^{K} \mathbf{w}_k \odot \mathbf{c}^{(k)}\right) \tag{28}$$

where $\mathbf{w}_k$ is the kth weight vector in $\mathbf{W}_{\text{hops}}$. This adaptive weighting dynamically balances spectral fidelity and contextual support: 0-hop features ($\mathbf{c}^{(0)}$) dominate when the center pixel is highly discriminative or context is noisy, while ≥1-hop features are emphasized in homogeneous regions.

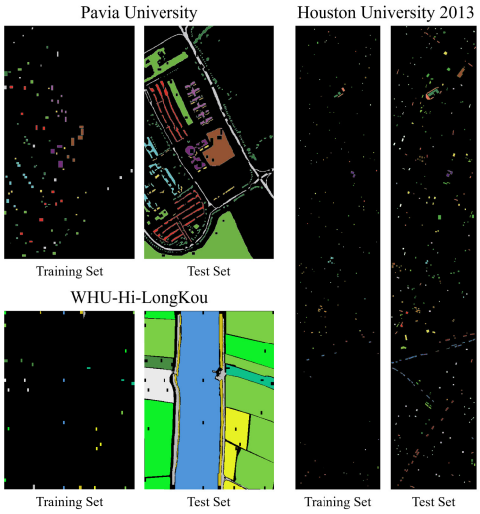


Fig. 2. Distribution of training and test samples on the three datasets.

A learnable temporal attention vector $\boldsymbol{\theta} \in \mathbb{R}^T$ is softmax-normalized to obtain timestepwise weights $\boldsymbol{\alpha}$

$$\boldsymbol{\alpha} = \text{Softmax}\left(\boldsymbol{\theta}\right). \tag{29}$$

The final center-node representation $\mathbf{z}_{\text{center}} \in \mathbb{R}^D$ is the weighted sum across the sequence dimension

$$\mathbf{z}_{\text{center}} = \sum_{t=1}^{T} \alpha_t \cdot \mathbf{C}_{\text{fused}}^{(t)}. \tag{30}$$

CSTA thus integrates adaptive multihop spatial aggregation with learnable temporal fusion to produce an enriched spatiotemporal feature $\mathbf{z}_{\text{center}}$, which is subsequently fed into the classification head to generate the output for the center pixel.

IV. Experiments

In this section, we present a series of experiments based on MS2GCAN to discuss the impact of certain parameters on the network and validate the method's effectiveness.

A. Dataset Description

Experiments were conducted on three widely used HSIC datasets: Pavia University (PU), Houston University (HU) 2013, and WHU-Hi-LongKou (WHLK).

1) Pavia University: PU, collected over Pavia, Italy, with the ROSIS sensor, contains 610 × 340 pixels, 103 spectral bands, and nine land-cover classes totaling 43 923 labeled samples. Spatial resolution is 1.3 m.

2) HU 2013: HU, acquired over the University of Houston, USA, using the CASI sensor [57], has 349 × 1905 pixels, 144 spectral bands, and 15 land-cover classes with 15 029 labeled samples. Spatial resolution is 2.5 m.

3) WHU-Hi-LongKou: WHLK, captured over Longkou, China, via a UAV-mounted Headwall Nano-Hyperspec sensor [58], contains 550 × 400 pixels, 270 spectral bands, and nine land-cover classes totaling 204 542 labeled samples. Spatial resolution is approximately 0.463 m.

Experiments on PU and HU follow the disjoint sampling strategy of [55]. For WHLK, 200 samples per class were

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES ON THE THREE DATASETS

NO.	Pavia University			Houston University 2013			WHU-Hi-LongKou		
	Class	Training	Test	Class	Training	Test	Class	Training	Test
1	Asphalt	548	6304	Healthy Grass	198	1053	Corn	200	34311
2	Meadows	540	18146	Stressed Grass	190	1064	Cotton	200	8174
3	Gravel	392	1815	Synthetic Grass	192	505	Sesame	200	2831
4	Trees	524	2912	Tree	188	1056	Broad-leaf soybean	200	63012
5	Metal Sheets	265	1113	Soil	186	1056	Narrow-leaf soybean	200	3951
6	Bare Soil	532	4572	Water	182	143	Rice	200	11654
7	Bitumen	375	981	Residential	196	1072	Water	200	66856
8	Bricks	514	3364	Commercial	191	1053	Roads and houses	200	6924
9	Shadows	231	795	Road	193	1059	Mixed weed	200	5029
10				Highway	191	1036			
11				Railway	181	1054			
12				Parking Lot1	192	1041			
13				Parking Lot2	184	285			
14				Tennis Court	181	247			
15				Running Track	187	473			
	Total	3921	40002	Total	2832	12197	Total	1800	202742

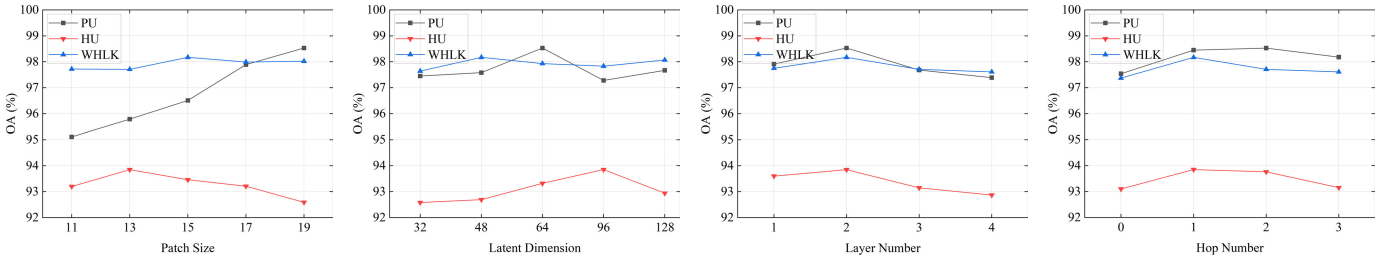


Fig. 3. Parameter analysis of MS2GCAN. Parameter analysis of different (a) patch sizes, (b) latent dimensions, (c) GCN layer numbers, and (d) hop numbers.

used for training, with the remainder for testing. The sample distributions are illustrated in Fig. 2, and detailed training/test counts are provided in Table I.

B. Experimental Setup

MS2GCAN was implemented in Python 3.9.23 and PyTorch 2.6.0 on Ubuntu, running on an Intel Xeon E5-2680 v4 CPU and Nvidia RTX 3090 GPU. LIF spiking neurons and the surrogate gradient method from SpikingJelly [56] 0.0.0.14 were used, with multistep training ($\beta = 0.5$, $\theta = 1.0$, and hard reset to 0).

Training employed Adam with an initial learning rate of $1e^{-4}$, cosine-annealed to $1e^{-6}$ over 200 epochs. Batch sizes were 64 for PU and 32 for HU/WHLK. Input patch sizes were 19, 13, and 15, and latent dimensions 64, 96, and 48 for PU, HU, and WHLK, respectively. The number of GCN layers $N = 2$, center-node hops $K = 2, 1, 1$, and timesteps $T = 3$. Models with the lowest training loss were used for evaluation.

Evaluation metrics include overall accuracy (OA), average accuracy (AA), Kappa (κ), and per-class accuracy. OA is the proportion of correctly classified pixels, AA is the mean per-class accuracy, and κ measures agreement beyond chance, particularly useful for imbalanced datasets.

C. Parameter Analysis

We analyze the sensitivity of MS2GCAN to key hyperparameters, including patch size, latent dimension, GCN depth, center-node neighborhood range, and the firing threshold θ, with results shown in Figs. 3 and 4.

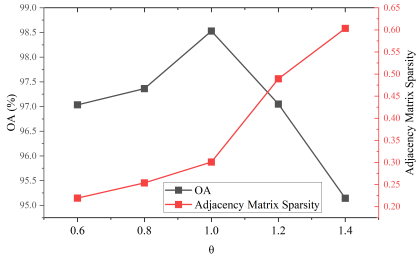


Fig. 4. Impact of the firing threshold θ on OA and adaptive graph sparsity on the PU dataset.

1) Patch Size: The optimal size reflects the ground sampling distance (GSD). For PU (1.3-m resolution), a 19×19 patch ($\sim 24.7 \times 24.7$ m^2) is needed to capture the spatial consistency of large homogeneous regions like "Meadows." HU (2.5-m resolution) reaches its peak at 13×13, covering the largest physical area ($\sim 32.5 \times 32.5$ m^2) to encapsulate expansive urban structures (e.g., commercial buildings). WHLK (0.463-m resolution) favors a 15×15 patch ($\sim 7 \times 7$ m^2); its performance remains stable across scales because the fine-grained, repetitive crop rows are effectively captured within a relatively small physical footprint.

2) Latent Dimension: The preferred latent dimension is dataset-dependent. PU achieves optimal performance with a medium-sized feature space, HU benefits from

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TABLE II
ABLATION STUDY OF THE SPIKING BACKBONE AND TEMPORAL DYNAMICS (OA%)

	ANN	SNN	Timestep	PU	HU	WHLK
1	✓		-	96.76	91.88	97.51
2		✓	1	96.15	93.35	97.02
3		✓	3	**98.53**	**93.85**	**98.17**
4		✓	5	98.23	93.42	97.98
5		✓	7	97.83	93.34	97.94
6		✓	9	97.40	93.54	97.91

higher-dimensional embeddings to model subtle spectral variations, whereas WHLK favors a more compact representation to avoid overfitting. Overall, MS2GCAN maintains stable performance across a wide range of feature dimensions.

3) GCN Depth: Two graph convolution layers consistently provide the best tradeoff between non-Euclidean contextual aggregation and feature discrimination. Shallower settings are insufficient for spatial modeling, while deeper architectures suffer from oversmoothing, confirming that a shallow GCN is adequate for HSIC.

4) Neighborhood Hops: The hop parameter defines the semantic reach in non-Euclidean space. PU benefits from a two-hop range, allowing the model to bridge spatially disjoint but semantically identical objects (e.g., fragmented shadows) across distances >20 m. Conversely, HU and WHLK favor one-hop, as their discriminative features are concentrated within the immediate structural neighborhood; higher-order aggregation in these fragmented or high-precision scenes would introduce spectral interference and "oversmoothing."

5) Firing Threshold θ: The firing threshold θ is a pivotal hyperparameter that dictates the spike density, which in turn influences the construction of the adaptive adjacency matrix. As shown in Fig. 4, increasing θ from 0.6 to 1.4 on the PU dataset leads to a monotonic increase in the adjacency matrix sparsity, rising from 21.94% to 60.35%. This confirms that θ effectively serves as a density regulator, filtering out redundant relational information by restricting spike generation. The OA reaches its peak at $\theta = 1.0$, whereas excessive sparsity (e.g., at $\theta = 1.4$) results in a performance dip, likely due to the loss of essential topological connectivity.

Overall, these results demonstrate that MS2GCAN is robust to hyperparameter variations and can flexibly adapt its spatial and spectral modeling capacity to diverse hyperspectral scenes.

D. Ablation Experiment

1) Efficacy of Spiking Backbone and Sensitivity to Timesteps: To evaluate the efficacy of the spiking backbone, we compare it with a single-step ANN baseline by replacing LIF neurons with ReLU and disabling temporal accumulation. As shown in Table II, the single-step SNN ($T = 1$) does not outperform the ANN, indicating that spiking activation alone provides limited benefit without temporal integration.

Enabling multistep processing leads to consistent performance gains, with the best accuracy achieved at $T = 3$. To further investigate the tradeoff between inference latency and accuracy, we illustrate the impact of T in Fig. 5. From the perspective of neuronal dynamics, a larger T allows for more sufficient accumulation of evidence in LIF neurons, enhancing the stability of spectral–spatial representations. However, as shown in Fig. 5, while OA improves significantly from $T = 1$ to $T = 3$, the accuracy gains marginalize or slightly degrade thereafter ($T > 3$), likely due to the oversmoothing of spiking responses.

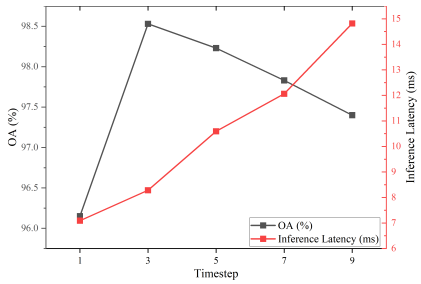


Fig. 5. OA and inference latency versus time step T on the PU dataset.

TABLE III
ABLATION STUDY ON THE PROPOSED MODULES (OA%)

	MSTE	ASGC	CSTA	PU	HU	WHLK
1				85.28	88.74	92.60
2	✓			90.23	90.78	93.76
3		✓		89.54	90.15	94.16
4	✓	✓		94.79	91.99	95.09
5		✓	✓	96.81	92.23	96.86
6	✓	✓	✓	**98.53**	**93.85**	**98.17**

Meanwhile, inference latency increases near-linearly with T because of the iterative nature of SNNs. Taking both factors into account, $T = 3$ is selected as the optimal balance point, providing state-of-the-art (SOTA) accuracy while maintaining low latency, which is critical for resource-constrained remote sensing applications.

2) Efficacy of the Proposed Modules: The baseline corresponds to a minimal SNN classifier composed of the STM, a spatiotemporal global pooling layer, and a classification head. The effectiveness and complementarity of the proposed MSTE, ASGC, and CSTA modules are evaluated by progressively activating MSTE and ASGC, and by substituting the global pooling operation with the proposed CSTA module, as summarized in Table III.

Introducing MSTE consistently improves performance across all three datasets, with particularly notable gains on PU and HU. This indicates that adaptive multiscale spatial feature extraction is crucial for handling land-cover regions with diverse spatial extents and structural variability.

When ASGC is incorporated on top of MSTE, further accuracy improvements are observed on all datasets, with more pronounced gains on PU and WHLK. This demonstrates that modeling feature-driven non-Euclidean relationships, rather than relying on fixed Euclidean neighborhoods, is beneficial for capturing complex spectral–spatial correlations in HSIs.

Replacing spatiotemporal global pooling with the CSTA module yields the most significant performance gain. Our analysis identifies multihop spatial fusion as the primary

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TABLE IV
COMPARATIVE STUDY OF GRAPH ADJACENCY MATRIX CONSTRUCTION METHODS (OA%)

	PU	HU	WHLK
4-NN	94.01	92.18	96.98
Euc-Gauss	97.81	92.67	97.62
Cos-Sim	97.92	92.63	97.68
APA	**98.53**	**93.85**	**98.17**

driver, as it extends the receptive field in non-Euclidean space to capture long-range structural correlations essential for complex land-cover discrimination. Furthermore, the learnable temporal weighting provides a supplementary enhancement (e.g., a 0.46% OA increase on PU). By adaptively emphasizing informative timesteps in the spike sequence, this component refines features and stabilizes the center-pixel representation against temporal noise. Together, these mechanisms enable CSTA to effectively integrate broad spatial context with fine-grained temporal dynamics for superior classification.

Finally, integrating all three modules achieves the best overall performance. These results confirm that MSTE, ASGC, and CSTA provide complementary benefits and together form a coherent framework for spatiotemporal HSIC.

3) Analysis of Adjacency Matrix Construction Methods: The comparison of adjacency construction methods in Table IV validates the effectiveness of the proposed APA strategy. While static four NN performs poorly due to its fixed spatial proximity, feature-driven metrics, like Euc–Gauss and Cos–Sim yield improvements by introducing adaptive modeling. Notably, APA outperforms Cos–Sim by explicitly incorporating feature centering before normalization. This design aligns with the motivation of mitigating membrane potential saturation, as centering filters out common-mode noise and background offsets that might otherwise dominate spiking activity. By decoupling structural correlations from absolute firing intensity, APA ensures that potential accumulation is driven by intrinsic spectral variations rather than energy levels. This stabilization of graph topology is particularly beneficial for spiking dynamics, enabling MS2GCAN to maintain high representational precision and achieve consistent performance gains across all datasets.

E. Comparison With Other Methods

To assess MS2GCAN, we compared it with representative SOTA models for HSIC: Transformers (MsET [59], MSST [60], GSC-ViT [31], and DSFormer [32]), GCNs (MRCAGCFN [54] and MS2FN [55]), and SNNs (SR-MDA [61], AFSA [44], and FRM-SNN [45]). All methods were evaluated using their optimal reported settings or official implementations.

1) Quantitative Analysis of Overall Performance: Tables V–VII show that MS2GCAN consistently achieves the highest OA, AA, and κ across all datasets, reaching 98.53% on PU, 93.85% on HU, and 98.17% on WHLK. MS2GCAN outperforms the second-best competitor by 1.66%–2.25% in OA, demonstrating robust generalization across urban scenes (PU and HU) and large-scale agricultural environments (WHLK) with distinct spatial structures and spectral characteristics.

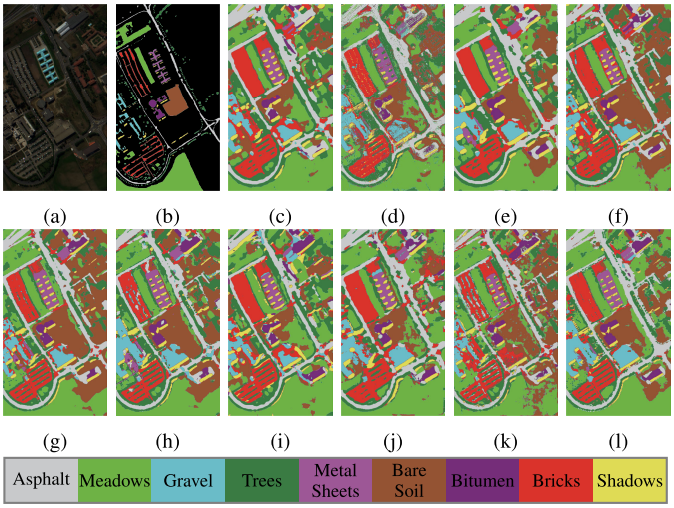


Fig. 6. Classification maps on the PU. (a) False-color image. (b) Ground-truth map. (c) MsET. (d) MSST. (e) GSC-ViT. (f) DSFormer. (g) MRCAGCFN. (h) MS2FN. (i) SR-MDA. (j) AFSA. (k) FRM-SNN. (l) MS2GCAN.

Compared with GCN-based methods, such as MS2FN, the observed improvements highlight the effectiveness of ASGC, where the APA enables feature-dependent, non-Euclidean relational modeling rather than relying on fixed spatial proximity. By constructing dynamic graph connections directly from spiking feature responses, ASGC provides more discriminative spatial associations for subsequent aggregation.

In contrast to Transformer-based approaches, which rely on global attention and may weaken local spatial consistency, MS2GCAN preserves local structures through MSTE while incorporating broader contextual cues via the proposed CSTA module. Notably, CSTA aggregates spatiotemporal information in a center-oriented manner with learnable hopwise spatial weighting and spike-driven temporal weighting. This design enables fine-grained spatiotemporal fusion while maintaining a compact and stable aggregation process.

Moreover, the clear margin over two SNN baselines indicates that temporal spiking dynamics alone are insufficient for high-accuracy HSIC. By jointly incorporating MSTE and adaptive non-Euclidean graph modeling via ASGC, together with center-oriented spatiotemporal aggregation in CSTA, MS2GCAN effectively enhances spectral–spatial representation learning in the spiking domain. This synergistic design enables MS2GCAN to achieve SOTA performance with a compact parameterization and efficient spike-driven computation.

At the class level, MS2GCAN achieves superior accuracy on most categories across all datasets. In particular, it performs consistently well on spectrally similar or structurally complex classes, such as urban materials in PU and transportation-related categories in HU. This observation suggests that jointly leveraging adaptive multiscale spatial encoding, feature-dependent non-Euclidean graph modeling, and center-focused spatiotemporal aggregation is beneficial for fine-grained class discrimination.

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TABLE V
CLASSIFICATION RESULTS ON THE PU

Class NO.	MsET	MSST	GSC-ViT	DSFormer	MRCAGCFN	MS2FN	SR-MDA	AFSA	FRN-SNN	MS2GCAN
1	90.78	91.53	91.72	88.25	95.16	91.59	86.84	90.37	86.96	**97.97**
2	98.19	95.27	97.72	92.95	96.87	98.40	94.17	97.51	93.60	**98.77**
3	74.55	85.51	90.85	80.39	68.32	**98.57**	66.06	81.54	61.16	97.18
4	96.26	93.65	92.00	96.15	97.42	98.63	97.41	79.70	94.09	**99.24**
5	99.55	99.46	99.10	**99.91**	99.46	99.46	99.10	98.20	99.82	**99.91**
6	63.15	75.77	99.04	99.28	**99.32**	95.98	54.06	79.44	93.44	97.64
7	87.26	91.13	**99.90**	97.35	99.29	94.70	79.19	91.23	94.39	99.69
8	98.69	96.88	96.88	95.39	98.72	96.61	97.73	98.87	98.57	**98.98**
9	95.60	84.40	93.58	96.98	98.62	**98.74**	97.23	92.20	95.97	97.91
OA (%)	91.57	91.82	96.14	93.18	95.95	96.87	86.36	92.17	91.76	**98.53**
AA (%)	89.34	90.40	95.64	94.07	94.80	96.96	85.76	89.90	90.89	**98.59**
$\kappa \times 100$	88.43	88.94	94.80	90.94	94.56	95.79	82.95	89.35	88.98	**98.02**

TABLE VI
CLASSIFICATION RESULTS ON THE HU 2013

Class NO.	MsET	MSST	GSC-ViT	DSFormer	MRCAGCFN	MS2FN	SR-MDA	AFSA	FRN-SNN	MS2GCAN
1	79.96	81.86	79.58	80.91	83.00	82.54	80.63	80.44	80.72	**83.04**
2	94.74	99.91	91.92	**100.00**	99.62	99.27	92.18	98.12	98.12	99.77
3	93.27	**100.00**	98.02	96.63	96.24	99.78	90.50	94.06	79.60	**100.00**
4	96.02	99.24	93.28	97.82	**99.91**	98.88	95.17	88.92	96.88	98.31
5	96.88	99.91	**100.00**	**100.00**	**100.00**	99.35	99.24	**100.00**	**100.00**	99.92
6	86.01	95.80	95.80	**100.00**	95.10	95.31	77.62	**100.00**	71.33	95.80
7	**95.24**	86.75	82.18	88.90	82.65	84.90	81.34	68.84	78.36	84.35
8	89.36	82.24	84.14	**95.82**	92.88	86.44	71.51	92.88	80.72	91.64
9	**90.56**	83.76	86.59	90.08	79.89	83.68	73.38	90.27	87.82	87.50
10	67.57	85.62	92.47	71.53	77.51	83.21	71.72	89.00	88.80	**97.92**
11	90.70	88.61	**99.81**	**99.81**	96.20	96.96	93.93	93.55	93.17	98.75
12	**96.45**	83.19	89.53	88.57	89.24	92.76	73.34	84.92	91.55	90.97
13	**91.58**	85.61	87.37	90.53	84.21	83.30	64.91	68.42	76.49	91.23
14	99.60	**100.00**	**100.00**	98.79	95.55	99.72	95.93	**100.00**	93.12	**100.00**
15	**100.00**	**100.00**	99.15	**100.00**	92.18	99.98	53.07	89.64	98.10	99.37
OA (%)	90.53	90.22	90.84	92.17	90.47	91.59	81.98	88.83	89.07	**93.85**
AA (%)	91.20	91.50	91.99	93.29	90.95	92.41	80.96	89.27	87.65	**94.57**
$\kappa \times 100$	89.72	89.38	90.07	91.50	89.65	90.87	80.67	87.87	88.14	**93.32**

TABLE VII
CLASSIFICATION RESULTS ON THE WHLK

Class NO.	MsET	MSST	GSC-ViT	DSFormer	MRCAGCFN	MS2FN	SR-MDA	AFSA	FRN-SNN	MS2GCAN
1	94.77	96.51	98.84	**99.80**	99.58	97.80	98.27	98.02	95.56	99.73
2	89.28	85.15	96.36	**99.02**	94.41	94.47	94.83	96.92	95.64	98.50
3	94.21	91.38	93.91	**99.54**	97.49	98.23	95.30	99.33	91.70	99.37
4	87.01	81.93	91.44	87.55	91.19	91.57	91.31	92.92	86.45	**95.77**
5	82.89	84.18	87.85	93.87	91.70	90.18	67.68	84.38	83.22	**96.24**
6	95.82	92.24	98.00	99.30	99.18	98.79	91.31	96.67	95.69	**99.75**
7	96.81	99.97	99.37	99.65	**99.99**	99.96	99.64	94.74	99.88	99.98
8	76.03	74.96	86.08	83.65	78.32	89.98	77.62	35.11	94.28	**94.30**
9	90.63	79.70	83.37	93.02	**96.06**	90.85	85.41	93.96	84.89	95.35
OA (%)	91.89	90.95	95.47	95.04	95.88	95.92	91.75	92.73	93.56	**98.17**
AA (%)	89.72	87.33	92.80	95.04	94.21	94.65	89.04	88.00	91.92	**97.67**
$\kappa \times 100$	89.49	88.30	94.08	93.56	94.62	94.66	92.57	90.54	91.63	**97.59**

2) Visual Analysis of Classification Maps: We provide qualitative comparisons on PU (see Fig. 6), HU (see Fig. 7), and WHLK (see Fig. 8). Overall, MS2GCAN produces classification maps with improved spatial consistency and clearer structural delineation compared with all baseline methods.

Convolution-driven SNN-based approaches exhibit noticeable interclass confusion (e.g., Meadows versus Bare Soil on PU), which can be attributed to their limited capability in modeling structured non-Euclidean spectral–spatial relations. By contrast, MS2GCAN benefits from ASGC's feature-dependent dynamic graph aggregation, yielding more coherent regions and substantially reducing isolated misclassifications.

Transformer- and GCN-based baselines tend to blur thin or elongated structures (e.g., road and highway boundaries in HU), likely due to oversmoothing effects introduced by global

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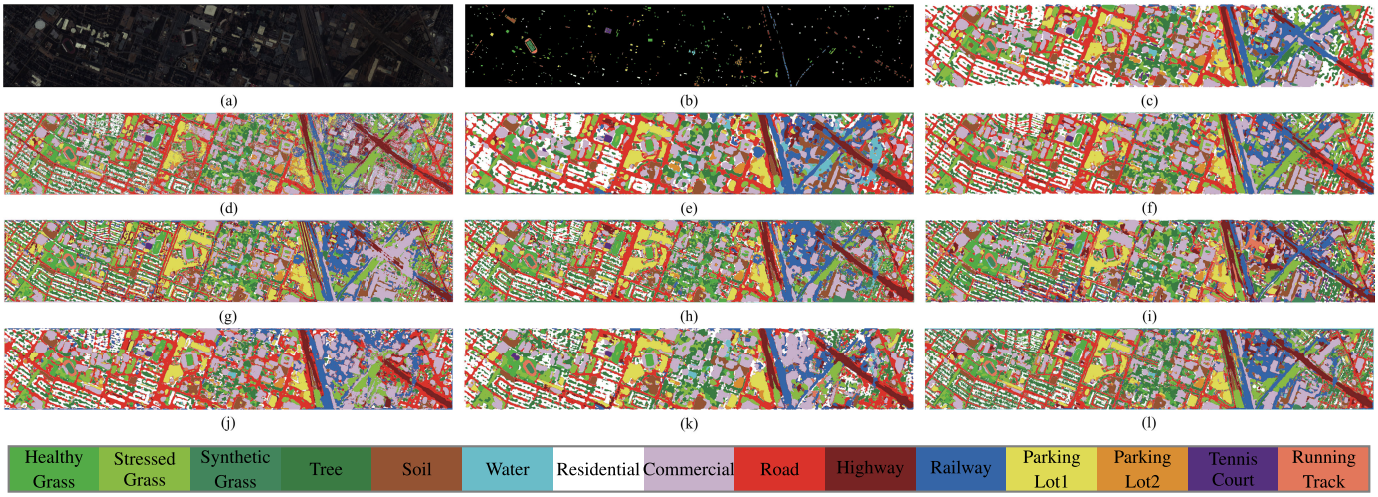


Fig. 7. Classification maps on the HU 2013. (a) False-color image. (b) Ground-truth map. (c) MsET. (d) MSST. (e) GSC-ViT. (f) DSFormer. (g) MRCAGCFN. (h) MS2FN. (i) SR-MDA. (j) AFSA. (k) FRM-SNN. (l) MS2GCAN.

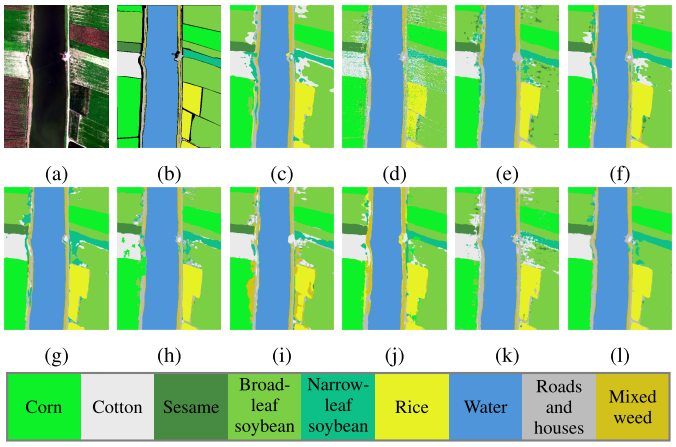


Fig. 8. Classification maps on the WHLK. (a) False-color image. (b) Ground-truth map. (c) MsET. (d) MSST. (e) GSC-ViT. (f) DSFormer. (g) MRCAGCFN. (h) MS2FN. (i) SR-MDA. (j) AFSA. (k) FRM-SNN. (l) MS2GCAN.

attention or uniform neighborhood aggregation. In comparison, MS2GCAN better preserves fine boundaries and structural continuity. On the WHLK dataset, it accurately delineates narrow crop-field borders, owing to the complementary effects of MSTE's multiscale spatial extraction and CSTA's center-oriented multihop aggregation, which jointly capture local geometry and broader contextual layout.

To quantitatively evaluate boundary preservation, we calculated the edge OA within a 3-pixel dilated buffer zone around Canny-extracted ground truth contours. As shown in Fig. 9, MS2GCAN achieves a superior edge OA of 98.12%, outperforming the Transformer-based baseline by 4.32%. The corresponding error maps further confirm that while Transformers suffer from oversmoothing due to global attention, MS2GCAN effectively maintains local geometric integrity and sharp transitions through adaptive graph aggregation.

Overall, the qualitative results are consistent with the quantitative improvements, demonstrating that MS2GCAN achieves

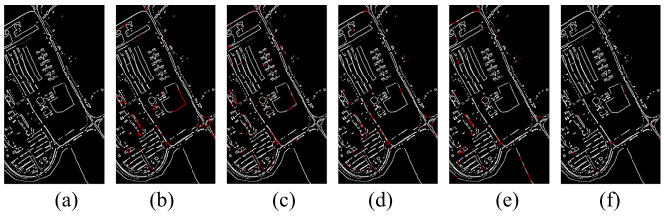


Fig. 9. Boundary preservation and error maps on PU dataset. Black: nonboundary background and white/red: correct/incorrect classification within the 3-pixel dilated zone. Edge OA is noted in subfigure titles. (a) Ground truth. (b) MsET edge OA = 91.09%. (c) MSST edge OA = 4 91.48%. (d) GSC-ViT edge OA = 93.80%. (e) DSFormer edge OA = 92.62%. (f) MS2GCAN edge OA = 98.12%.

a favorable balance between spatial smoothness and boundary fidelity, outperforming representative SNN-, GCN-, and Transformer-based methods.

F. Space Complexity: Parameters and Memory Footprint

The space complexity is evaluated via parameter volume and inference memory. While parameters are determined by the architecture, memory depends on the input scale. Thus, we standardize the input (patch size 15×15 and PCA dimension 30) and fix the inference batch size to 32 to evaluate peak demand.

As shown in Table VIII, MS2GCAN achieves the most compact parameterization, which is significantly fewer than Transformer-based MsET and SNN-based FRM-SNN. This results from the lightweight spiking backbone and the efficient design of the CSTA module. Under the standardized input, MS2GCAN maintains a competitive memory footprint. Although SNNs require additional memory for membrane potentials, the overall lightweight architecture remains suitable for resource-constrained platforms.

To further investigate the scalability of the APA matrix, which has an $O((L^2)^2)$ complexity, we evaluate the peak memory demand across varying patch sizes $L \in \{11, 15, 19, 23, 27\}$. As illustrated in Fig. 10, although the theoretical complexity is quadratic with respect to the node count (L^2), the actual

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TABLE VIII
COMPARISON OF MODEL COMPLEXITY AND MEMORY FOOTPRINT

Models	Parameters(M)			Inference Memory(GB)		
	PU	HU	WHLK	PU	HU	WHLK
MsET	2.29	2.29	2.29	0.84	0.77	1.26
MSST	0.42	0.42	0.42	0.18	**0.11**	0.60
GSCViT	0.29	0.29	0.29	0.18	0.24	0.73
DSFormer	0.68	0.68	0.68	0.24	0.26	0.26
MRCAGCFN	0.61	0.62	0.63	**0.11**	**0.11**	**0.11**
MS2FN	0.46	0.47	0.48	0.79	0.95	1.09
SR-MDA	1.71	1.71	1.71	1.51	0.59	6.84
AFSA	2.23	2.23	2.21	1.22	1.15	1.64
FRM-SNN	1.14	1.17	1.14	0.28	0.21	0.70
MS2GCAN	**0.08**	**0.18**	**0.06**	0.25	0.19	0.65

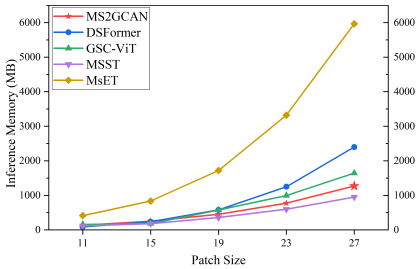


Fig. 10. Memory consumption versus patch size L for MS2GCAN and Transformer-based methods.

memory growth of MS2GCAN exhibits a significantly gentler slope compared to many Transformer-based counterparts. This is primarily due to the architectural efficiency in structural modeling: while Transformer-based models require storing large-scale, multilayered, and multiheaded attention maps as intermediate buffers, MS2GCAN generates the dynamic adjacency matrix directly from the spiking features, avoiding the cumulative memory overhead of deep feature-to-feature attention stacking. For instance, as L increases from 11 to 27, the inference memory of MS2GCAN grows by only about 1100 MB, maintaining a favorable scalability for spatial context modeling.

G. Energy Efficiency: Quantitative Analysis and Computational Cost

To provide a level playing field, we evaluate floating point operations (FLOPs) and synaptic operations (SOPs) under the same $15 \times 15 \times 30$ input. Following the 45-nm complementary metal–oxide–semiconductor (CMOS) energy model [62], we set $E_{MAC} = 4.6$ pJ and $E_{AC} = 0.9$ pJ. For ANN components (including the encoding layer), we calculate FLOPs. For SNN-based backbone components, SOPs are defined as follows:

$$SOPs = \sum_{k=1}^{K} (T \times fr_k \times \Phi_k) \tag{31}$$

where T, fr_k, and Φ_k denote the time steps, average firing rate, and equivalent ANN-layer FLOPs, respectively. The total energy E_{total} per sample is estimated as follows:

$$E_{total} = FLOPs \times E_{MAC} + SOPs \times E_{AC}. \tag{32}$$

TABLE IX
COMPARISON OF COMPUTATIONAL COST AND ENERGY EFFICIENCY

Models	FLOPs (M)	SOPs (M)	Energy (μJ)
MsET	7304.79	0.00	33602.02
MSST	91.32	0.00	420.07
GSCViT	90.03	0.00	414.14
DSFormer	55.53	0.00	255.44
MRCAGCFN	65.18	0.00	299.83
MS2FN	107.41	0.00	494.09
SR-MDA	4.99	158.59	165.68
AFSA	0.00	231.04	207.93
FRM-SNN	3.89	142.43	146.08
MS2GCAN	12.59	7.84	**64.96**

As per Table IX (evaluated on the PU dataset), MS2GCAN achieves the lowest energy of 64.96 μJ/sample, representing a reduction of over 50% compared to the second-best performer, FRM-SNN. In MS2GCAN, the MSTE and ASGC modules handle the majority of spectral–spatial encoding through sparse additions, while the final lightweight CSTA module provides precise feature aggregation. This combination maintains a balance between the low power consumption of spiking dynamics and the representational accuracy of the gating mechanism.

H. Influence of Different Numbers of Training Samples

The OA comparison under scarce training samples (see Fig. 11) demonstrates the strong generalization ability of MS2GCAN in data-limited settings. Although performance improves with more training samples, MS2GCAN consistently outperforms all competitors across the three datasets, with the largest margins appearing under extremely sparse supervision. This advantage can be attributed to the joint effect of MSTE, adaptive neighborhood learning (ASGC), and center-focused spatiotemporal aggregation (CSTA), which collectively enhance structural consistency and suppress noisy local responses. As a result, MS2GCAN is able to extract more discriminative and robust representations from limited labeled samples, making it particularly effective in realistic HSI scenarios where annotation is scarce and costly.

I. Robustness to Noise

To evaluate the model stability against sensor noise, Gaussian noise with standard deviation σ is added to the test sets. As shown in Fig. 12, the performance of DSFormer declines rapidly on all three datasets, and other Transformer- or GCN-based methods also show varying degrees of degradation. In contrast, SNN-based methods demonstrate remarkable robustness on PU and HU datasets. Notably, FRM-SNN exhibits a performance dip on the WHLK dataset, which could be attributed to its residual multibranch structure being more sensitive to noise in large homogeneous regions. MS2GCAN remains stable across all scenarios, benefiting from the nonlinear thresholding of spikes and the temporal integration mechanism which naturally suppress stochastic noise. Furthermore, the adaptive graph aggregation acts as a spatial low-pass filter, preserving the structural integrity of large homogeneous regions.

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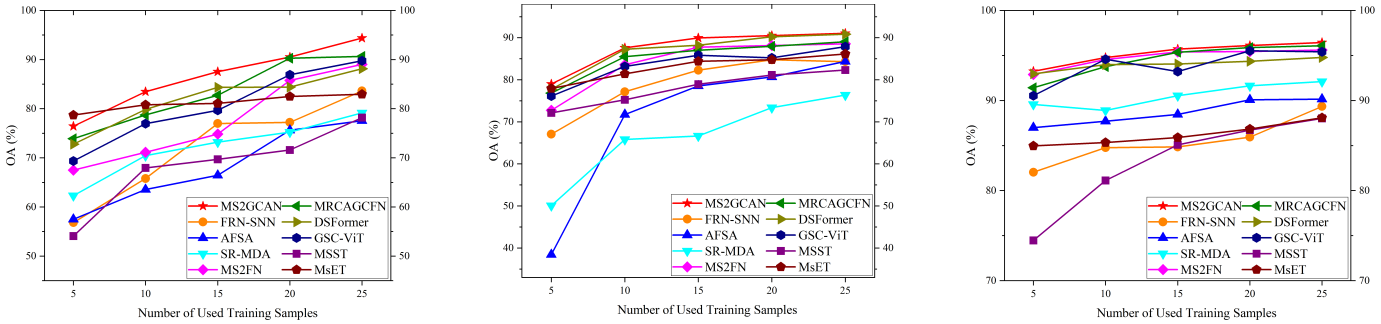


Fig. 11. OA (%) comparison of different models with varying numbers of training samples. Performance on (a) PU, (b) HU, and (c) WHLK datasets.

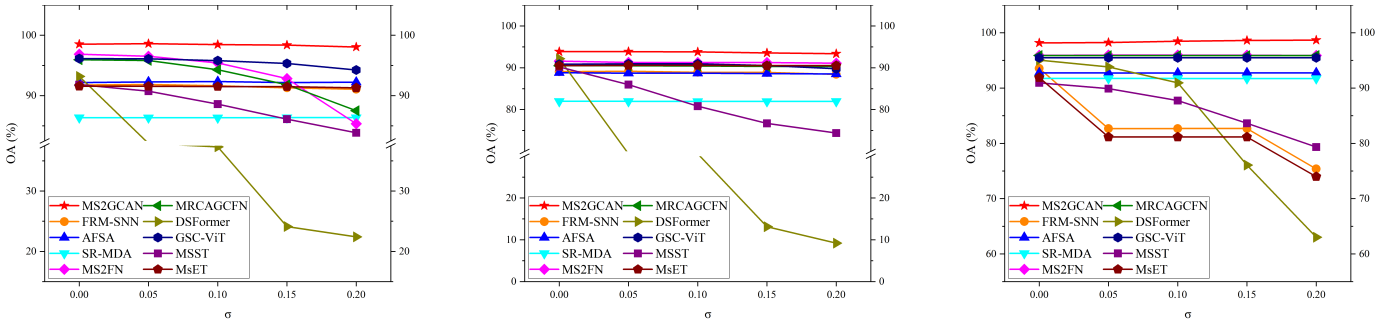


Fig. 12. OA (%) comparison of different models under varying standard deviations σ of Gaussian noise on (a) PU, (b) HU, and (c) WHLK datasets.

V. Conclusion

In this article, we introduced MS2GCAN, a compact framework for HSIC that integrates SNN temporal dynamics with adaptive graph-based non-Euclidean modeling. MS2GCAN incorporates three modules: MSTE for adaptive multiscale spatial encoding, ASGC for feature-dependent dynamic graph construction and spiking aggregation, and CSTA for multihop spatial fusion with learnable temporal weighting. Experiments on three benchmark datasets demonstrate that MS2GCAN consistently achieves SOTA performance with a reduced parameter budget. Ablation and efficiency analyses further validate the contribution of each module and the compactness of the framework. Future work will investigate neuromorphic deployment, extend the framework to dense prediction tasks, such as segmentation and change detection, and explore advanced spiking neuron models and graph sparsification.

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Shufang Xu (Member, IEEE) received the B.S. degree in applied electronic technology from Anhui Normal University, Wuhu, China, in 2002, and the master's and Ph.D. degrees in communication engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2005 and 2017, respectively.

She is currently an Associate Professor with the College of Computer Science and Software Engineering, Hohai University, Nanjing. In the last five years, she was participating in two research projects supported by the National Key Research and Development Program of China and one research project supported by Jiangsu Province Key Research and Development Program. Her research interests include UAV swarm cooperation, information fusion, and image processing in remote sensing.

Dr. Xu's part of the research results has been applied in water-resource flood-disaster monitoring and operation decision supporting system of water conservancy, which won China Industry-University-Research Cooperation Innovation Award in 2019 and 2021.



Wenxuan Fei received the B.E. degree in information and computing science from Suzhou University of Technology, Suzhou, China, in 2024. He is currently pursuing the M.S. degree in electronic and information engineering with the College of Computer Science and Software Engineering, Hohai University, Nanjing, China.

His research interests include deep learning and hyperspectral image processing.



Gaoxu Wang received the B.S. degree in resource and environmental regionalization and management and the M.S. degree in hydrology and water resources from Sun Yat-sen University, Guangzhou, China, in 2002 and 2005, respectively, and the Ph.D. degree in hydrology and water resources from Nanjing Hydraulic Research Institute, Nanjing, China, in 2012.

He is currently a Professor and the Deputy Director of the Big Data and Smart Water Conservancy Research Center, Nanjing Hydraulic Research Institute. His research interests include machine learning, neural networks, and image processing in remote sensing.



Runhua Sheng (Graduate Student Member, IEEE) received the B.E. degree in safety engineering from Zhengzhou University, Zhengzhou, China, in 2021, and the M.S. degree in computer science and technology from the College of Computer Science and Software Engineering, Hohai University, Nanjing, China, in 2025. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include image processing and computer vision.



Zhonghao Chen (Graduate Student Member, IEEE) received the M.S. degree (summa cum laude) from the College of Computer and Information, Hohai University, Nanjing, China, in 2022, where he is currently pursuing the Ph.D. degree with the College of Computer and Information.

His research interests include deep learning and hyperspectral image processing.



Hongmin Gao (Senior Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in computer application technology from Hohai University, Nanjing, China, in 2006 and 2014, respectively.

He is currently a Professor with the College of Computer Science and Software Engineering, Hohai University. In the last five years, he has been undertaking two research projects supported by the National Natural Science Foundation of China (NSFC), one research project supported by Jiangsu Natural Science Foundation, one research project supported by the transformation of scientific and technological achievements in Jiangsu Province, and participating in the National Key Research and Development Program of China. His research interests include deep learning, information fusion, and image processing in remote sensing.

Dr. Gao's part of the research results has been applied in water-resource flood-disaster monitoring and operation decision supporting system of water conservancy, which won the Second Prize of Jiangxi Province Science and Technology Advancement Reward in 2018.

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